

GSC-X Internet Infrastructure Sandbox

Internet Infrastructure Validation V1: Replay-Based Validation of Executive Decision Support, Digital Twin Prediction and Executive Benchmark Standard Using Public Infrastructure Datasets

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Patent-Pending Architecture Portfolio

PV 2026-216 (GSC-F)
PV 2026-236 (GSC-S)
PV 2026-241 (GSC-C)
PV 2026-242 (GSC-X)

July 3, 2026

Abstract

Modern Internet infrastructure generates massive volumes of operational telemetry originating from cloud platforms, distributed services, AI workloads and large-scale data centers. While conventional monitoring primarily detects infrastructure degradation after operational thresholds are exceeded, replay-based validation enables evaluation of predictive decision-support methods under fully reproducible conditions.

This report presents the **GSC-X Internet Infrastructure Sandbox**, a validation environment integrating replay processing, Executive-state evaluation, Digital Twin prediction,

What-if analysis and Executive Benchmark Standard (EBS) certification.

Validation was performed using Alibaba Public Infrastructure replay, Google Cluster replay pipeline verification and six synthetic infrastructure incident scenarios including traffic spikes, AI bursts, cache failures, network degradation, database slowdown and compound operational crises.

Replay validation was successfully completed for all investigated scenarios. Executive certification reached an overall score of 96.42 for the Alibaba Public Replay, while Digital Twin What-if analysis identified the Hybrid mitigation strategy as the best-performing configuration within the implemented simulation model.

The Internet Infrastructure Sandbox establishes a reproducible validation framework for evaluating predictive Executive decision support using public infrastructure datasets. The presented results represent replay-based validation and should not be interpreted as production deployment results.

1 Introduction

Cloud platforms, hyperscale data centers and distributed Internet services operate under continuously changing workloads that may evolve from normal operation to congestion within a short period of time. Conventional monitoring systems provide detailed operational metrics but frequently detect infrastructure degradation only after measurable service deterioration has already occurred.

Replay-based validation offers an alternative approach by evaluating historical infrastructure telemetry under fully reproducible conditions. Instead of interacting with production systems, replay enables repeated execution of identical operational sequences while comparing alternative monitoring, prediction and mitigation strategies.

The GSC-X Internet Infrastructure Sandbox was developed as the Internet Infrastructure validation domain of the broader GSC-X Platform. Its objective is not to replace existing orchestration technologies but to evaluate whether a common Executive decision layer can provide predictive operational awareness across heterogeneous infrastructure datasets.

The implemented framework combines Unified Telemetry conversion, Executive-state estimation, replay validation, Digital Twin prediction, What-if analysis and Executive Benchmark Standard certification into a single reproducible workflow.

Validation was performed using Alibaba Public Infrastructure telemetry, Google Cluster replay pipeline verification and six representative synthetic incident scenarios. Together these experiments evaluate replay execution, predictive recommendation generation and standardized Executive certification under reproducible conditions.

The principal contribution of this report is the demonstration of a unified validation framework capable of processing heterogeneous infrastructure telemetry while preserving one Executive architecture, one replay methodology and one Executive Benchmark Standard.

2 Validation Architecture

The Internet Infrastructure Sandbox follows a unified validation workflow used throughout the GSC-X Platform. Independent infrastructure datasets are first converted into a common Unified Telemetry schema before being processed by the Executive Core. The Executive Core continuously estimates operational state variables that are subsequently used for replay validation, Digital Twin prediction, What-if analysis and Executive Benchmark Standard (EBS) certification.

This architecture separates dataset-specific preprocessing from Executive decision logic. Consequently, new infrastructure datasets can be integrated by implementing only a telemetry adapter while preserving the same Executive Core and evaluation methodology.

Figure 1 illustrates the validation workflow implemented in Version 1 of the Internet Infrastructure Sandbox.

GSC-X Internet Infrastructure Sandbox Architecture

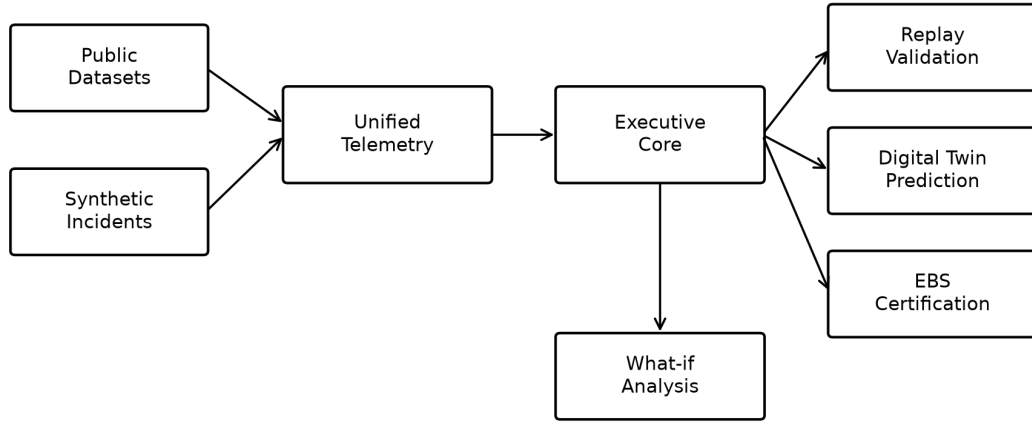


Figure 1: Validation workflow of the GSC-X Internet Infrastructure Sandbox. Public infrastructure datasets are converted into a Unified Telemetry representation before replay execution, Executive-state estimation, Digital Twin prediction and Executive Benchmark Standard certification.

Table 1 summarizes the principal validation components.

Table 1: Core components of the Internet Infrastructure Sandbox.

Component	Purpose
Unified Telemetry	Standardized infrastructure representation
Replay Engine	Deterministic telemetry replay
Executive Core	Continuous state estimation
Prediction Engine	Future stress estimation
Digital Twin	Evaluation of alternative strategies
What-if Engine	Comparative operational analysis
Executive Benchmark Standard	Unified performance evaluation
Executive Certification	Standardized executive scoring

The architecture was intentionally designed to remain independent of individual infrastructure providers, allowing the same validation pipeline to process multiple public datasets using identical Executive-state evaluation.

3 Validation Portfolio

Version 1 of the Internet Infrastructure Sandbox combines public infrastructure datasets, synthetic incident scenarios and Digital Twin simulations into a single reproducible validation framework.

The objective is to evaluate the Executive Core under heterogeneous operational conditions while preserving one replay methodology, one telemetry schema and one Executive Benchmark Standard.

Figure 2 summarizes the complete validation portfolio.

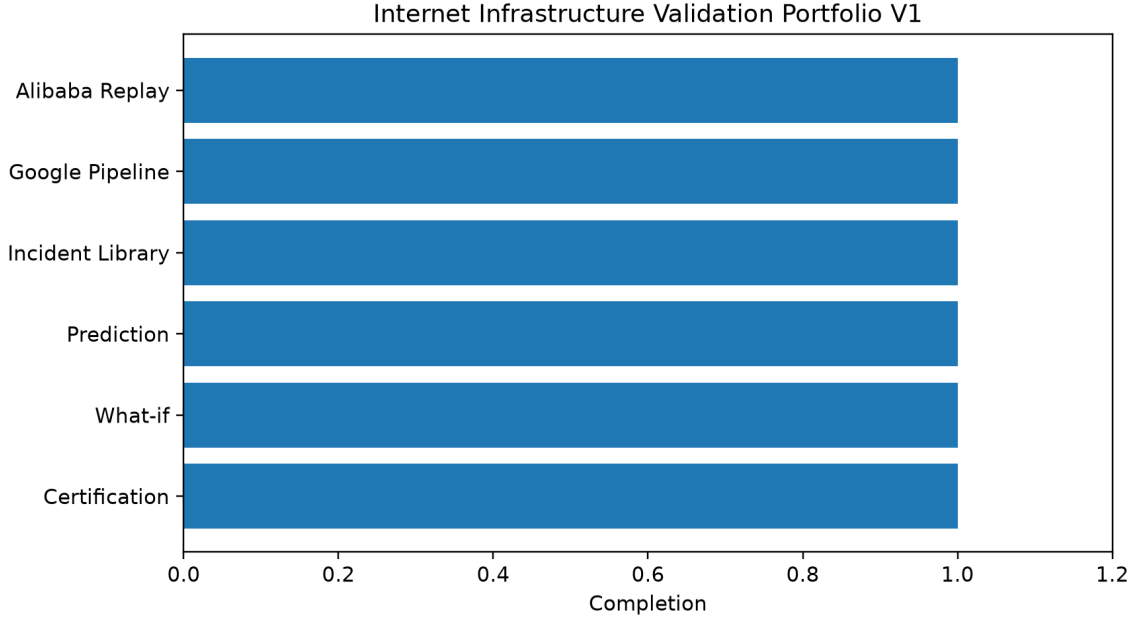


Figure 2: Validation portfolio implemented in the Internet Infrastructure Sandbox. The framework combines public replay datasets, synthetic infrastructure incidents, Digital Twin prediction and Executive Benchmark Standard certification.

Table 2 lists the completed validation stages.

Table 2: Validation portfolio completed in Version 1.

Validation	Dataset	Status
Public Replay	Alibaba Public Infrastructure	Complete
Replay Pipeline	Google Cluster Sample	Complete
Traffic Spike	Synthetic Incident	Complete
AI Burst	Synthetic Incident	Complete
Cache Failure	Synthetic Incident	Complete
Network Degradation	Synthetic Incident	Complete
Database Slowdown	Synthetic Incident	Complete
Compound Crisis	Synthetic Incident	Complete
Digital Twin Prediction	Executive Prediction Engine	Complete
What-if Analysis	Operational Simulation	Complete
Executive Certification	Executive Benchmark Standard	Complete

Together these validation stages establish the baseline reference implementation of the Internet Infrastructure Sandbox and provide reproducible evidence for replay validation, predictive Executive-state estimation and standardized Executive certification.

4 Alibaba Public Replay Validation

Alibaba Public Infrastructure was selected as the primary replay dataset for Version 1 of the Internet Infrastructure Sandbox.

The original telemetry was converted into the Unified Telemetry schema and processed by the Executive Core without dataset-specific modifications. Replay execution completed successfully for all samples while continuously estimating Executive-state variables and generating Executive certification metrics.

Figure 3 illustrates the replay validation workflow.

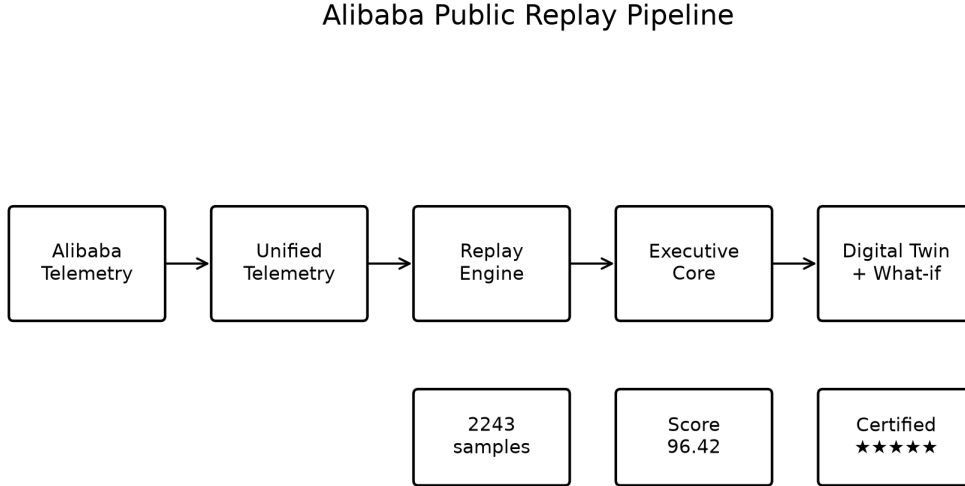


Figure 3: Alibaba Public Replay processed by the Internet Infrastructure Sandbox. Replay telemetry is converted into the Unified Telemetry schema before Executive-state evaluation and certification.

Table 3 summarizes the replay execution.

Table 3: Alibaba Public Replay summary.	
Property	Value
Dataset	Alibaba Public Infrastructure
Replay Samples	2243
Replay Status	Complete
Executive Core	Version 1.0
Digital Twin	Complete
Executive Certification	Complete

The completed replay demonstrates that heterogeneous public infrastructure telemetry can be processed using the common Executive Core while preserving deterministic replay execution and standardized Executive-state evaluation.

4.1 Executive Benchmark Standard Results

Following replay execution, the Alibaba dataset was evaluated using the Executive Benchmark Standard (EBS). The certification integrates replay stability, predictive capability, recovery behaviour and Executive intervention into a single Executive Score.

Table 4 summarizes the certification obtained during replay validation.

Table 4: Executive Benchmark Standard results for Alibaba Public Replay.

Metric	Value
Stability Score	100.00
Prediction Score	94.49
Recovery Score	93.45
Intervention Score	95.52
Reliability Score	96.69
Overall Executive Score	96.42
Certification Grade	Five-Star

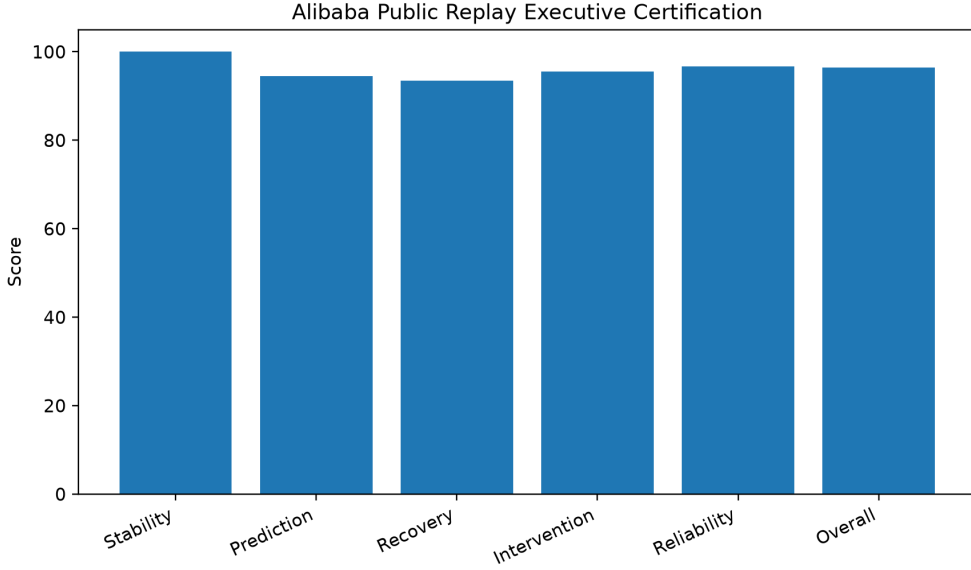


Figure 4: Executive Benchmark Standard certification obtained for the Alibaba Public Replay. The overall Executive Score reached 96.42, with consistently high scores across stability, prediction, recovery and intervention metrics.

The Alibaba Public Replay achieved an overall Executive Score of 96.42, establishing the reference baseline for subsequent Internet Infrastructure validations. Throughout replay execution the Executive Core remained stable and successfully completed prediction, Digital Twin evaluation and certification without interruption.

5 Google Cluster Replay Validation

To verify interoperability across independent infrastructure datasets, the replay pipeline was additionally validated using representative Google Cluster telemetry.

Unlike the Alibaba replay, the objective of this experiment was pipeline verification rather than large-scale benchmarking. The Google adapter successfully converted telemetry into the Unified Telemetry schema and executed the complete replay workflow using the same Executive Core.

Figure 5 illustrates the Google replay workflow.

Google Cluster Replay Pipeline Verification

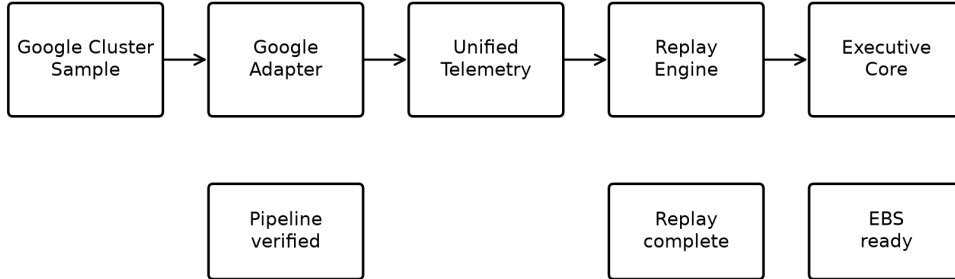


Figure 5: Google Cluster replay validation. The same Executive Core processes telemetry after Unified Telemetry conversion, demonstrating dataset-independent replay capability.

Table 5 summarizes the completed validation.

Table 5: Google Cluster replay summary.	
Property	Value
Dataset	Google Cluster Sample
Replay Pipeline	Complete
Telemetry Conversion	Complete
Executive Core	Version 1.0
Replay Status	Complete
Certification Pipeline	Complete

The successful replay confirms that the Internet Infrastructure Sandbox can process multiple public infrastructure datasets using the same validation workflow. The current implementation establishes technical readiness for future large-scale Google Cluster replay experiments.

6 Public Incident Library

Replay datasets capture historical infrastructure behaviour but cannot represent every operational situation. To extend validation beyond recorded telemetry, the Internet Infrastructure Sandbox includes a Public Incident Library consisting of reproducible synthetic infrastructure scenarios.

Version 1 implements six representative operational incidents covering computational overload, communication degradation and compound infrastructure failures.

Table 6 summarizes the implemented scenarios.

Table 6: Public Incident Library.

Scenario	Description
Traffic Spike	Sudden traffic increase
AI Burst	High computational workload
Cache Failure	Cache degradation
Network Degradation	Increased communication latency
Database Slowdown	Backend performance degradation
Compound Crisis	Combined multi-factor disturbance

Each scenario was executed using the identical replay methodology and Executive Core, allowing direct comparison between different classes of infrastructure disturbances.

6.1 Incident Validation Results

All incident scenarios were evaluated using the same replay engine, Executive Core and Executive Benchmark Standard. This ensured that the resulting Executive Scores reflected differences in infrastructure behaviour rather than changes in the evaluation methodology.

Figure 6 compares the Executive Scores obtained for all investigated scenarios.

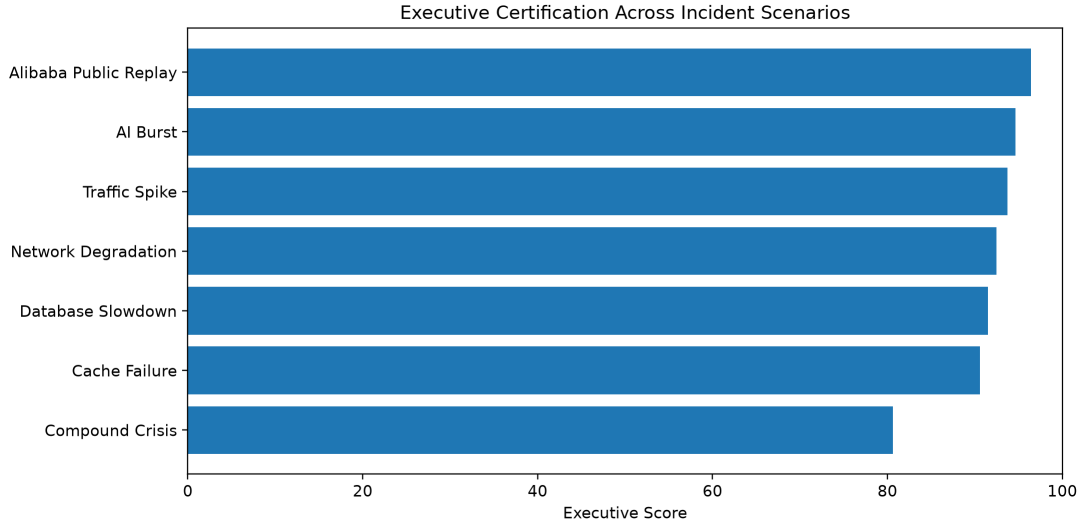


Figure 6: Executive Benchmark Standard scores for all replay and synthetic incident scenarios.

Table 7 summarizes the certification results.

Table 7: Executive certification across incident scenarios.

Scenario	Executive Score	Grade
Alibaba Public Replay	96.42	
AI Burst	94.65	
Traffic Spike	93.70	
Network Degradation	92.49	
Database Slowdown	91.51	
Cache Failure	90.56	
Compound Crisis	80.64	

All scenarios completed successfully without interruption of replay execution. The Executive

Score decreased as incident complexity increased, with the Compound Crisis representing the most demanding validation scenario in Version 1.

7 Digital Twin Prediction

The Digital Twin extends replay validation by estimating future infrastructure behaviour from the continuously evolving Executive state.

For each replay step the Prediction Engine estimates future Digital Stress, expected latency and a recommended Executive action. These predictions are subsequently evaluated using the What-if Engine before any mitigation strategy is selected.

Table 8 summarizes the prediction results.

Table 8: Digital Twin prediction summary.	
Property	Value
Prediction Samples	2231
First CACHE_BOOST	Timestamp 897
First PRE_SCALE	Timestamp 966
Maximum Predicted Stress	0.8428
Maximum Predicted Latency	1029.96 ms

The Prediction Engine generated Executive recommendations throughout the replay while continuously estimating future infrastructure conditions.

7.1 Recommended Executive Actions

Three recommendation categories were generated during prediction.

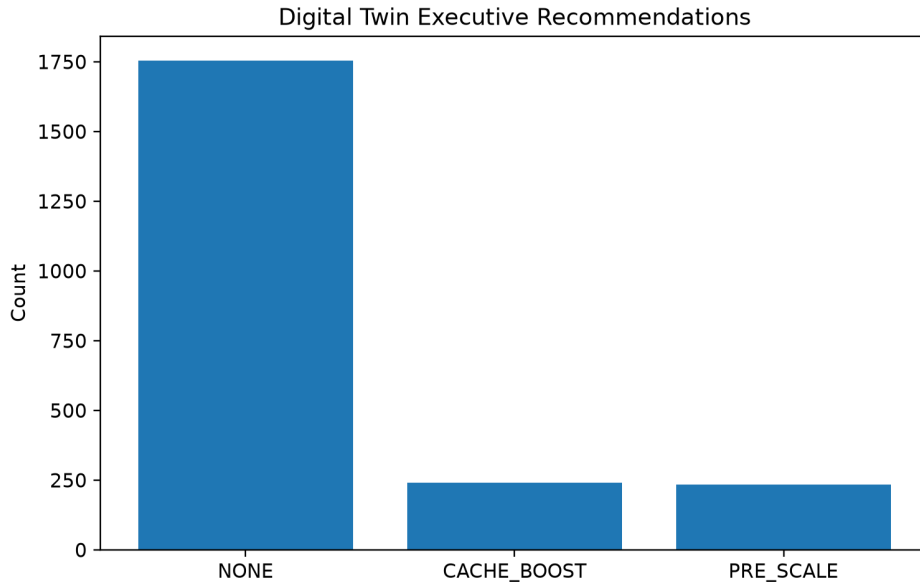


Figure 7: Distribution of Executive recommendations generated during Digital Twin prediction. Most replay samples required no intervention, while CACHE_BOOST and PRE_SCALE recommendations were issued during periods of increasing Digital Stress.

Table 9: Executive recommendations generated by the Prediction Engine.

Recommendation	Count
NONE	1754
CACHE_BOOST	242
PRE_SCALE	235

Most replay samples required no intervention, while predictive recommendations were generated during periods of increasing Digital Stress. CACHE_BOOST was typically recommended earlier in the replay, followed by PRE_SCALE during more demanding infrastructure conditions.

8 What-if Analysis

The What-if Engine evaluates alternative mitigation strategies using identical replay telemetry. Five representative operational strategies were investigated.

Figure 8 compares the resulting latency reduction.

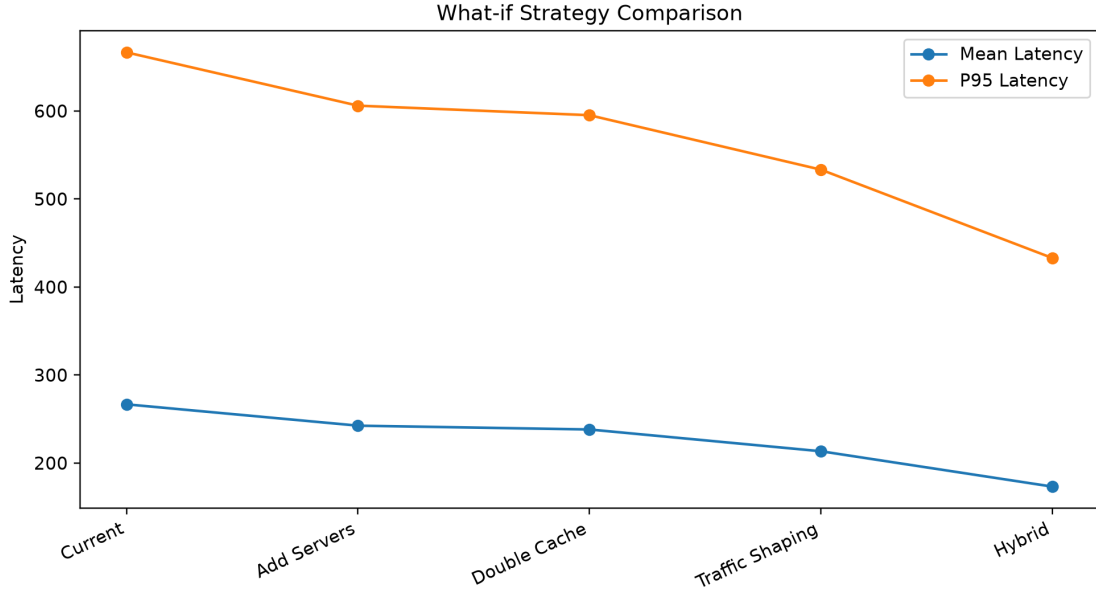


Figure 8: Comparison of the evaluated What-if strategies.

Table 10 summarizes the measured performance.

Table 10: What-if analysis results.

Strategy	Mean Latency	P95	Queue	Energy
Current	266.73	666.64	232.89	0.09331
Add Servers	242.48	606.04	211.72	0.09359
Double Cache	238.15	595.22	232.89	0.09331
Traffic Shaping	213.39	533.31	186.31	0.09331
Hybrid	173.20	432.88	169.37	0.09359

Within the implemented simulation model, the Hybrid strategy achieved the lowest average latency and the lowest 95th percentile latency while maintaining energy consumption comparable

to the remaining evaluated strategies.

9 Discussion

The completed validation demonstrates that replay-based infrastructure telemetry can be processed using a common Executive validation framework across heterogeneous datasets.

Alibaba Public Replay established the reference benchmark for Version 1, while Google Cluster replay verified interoperability of the adapter architecture. Synthetic incident scenarios further extended the evaluation by exposing the Executive Core to representative infrastructure disturbances ranging from isolated workload spikes to compound operational crises.

The Digital Twin and What-if analyses demonstrated that replay validation can be naturally extended toward predictive operational planning. Within the implemented simulation model, the Hybrid strategy consistently produced the lowest latency while maintaining comparable energy consumption.

Rather than introducing another infrastructure monitoring platform, the Internet Infrastructure Sandbox demonstrates a reproducible methodology for combining replay validation, Executive-state estimation, predictive analysis and standardized Executive certification within a single validation workflow.

10 Limitations

Several limitations should be considered when interpreting the presented results.

The reported experiments were performed using replay validation rather than live production infrastructure. Consequently, Executive recommendations were evaluated within deterministic replay scenarios and not through direct interaction with operational cloud systems.

The Google Cluster validation focused on replay pipeline verification using representative samples and should not be interpreted as a complete benchmark of the full Google Cluster dataset.

Similarly, the implemented Digital Twin represents a simulation model intended for comparative analysis of operational strategies. Although useful for reproducible evaluation, additional industrial validation would be required before deployment in production environments.

11 Conclusion

This report presented Version 1 of the GSC-X Internet Infrastructure Sandbox.

The completed validation integrates replay processing, Unified Telemetry conversion, Executive-state estimation, Digital Twin prediction, What-if analysis and Executive Benchmark Standard certification into a single reproducible framework.

Validation was successfully completed using Alibaba Public Infrastructure replay, Google Cluster replay pipeline verification and six representative infrastructure incident scenarios. All validation stages employed the same Executive Core architecture, enabling consistent evaluation across heterogeneous datasets.

The resulting Executive certification, together with Digital Twin prediction and What-if analysis, establishes the Internet Infrastructure Sandbox as the Internet Infrastructure validation domain of the broader GSC-X Platform.

Future development will focus on expanding validation using additional public datasets, cloud-native telemetry and industrial partner data while preserving the common Executive architecture and validation methodology.

12 Validation Summary

Table 11 summarizes the principal validation results obtained in Version 1 of the Internet Infrastructure Sandbox.

Table 11: Summary of completed validation activities.

Validation Activity	Result	Status
Alibaba Public Replay	Successful	Complete
Google Replay Pipeline	Successful	Complete
Unified Telemetry	Verified	Complete
Replay Engine	Verified	Complete
Executive Core	Verified	Complete
Prediction Engine	Verified	Complete
Digital Twin	Verified	Complete
What-if Analysis	Verified	Complete
Executive Benchmark Standard	Verified	Complete
Executive Certification	Completed	Complete

The Internet Infrastructure Sandbox successfully completed all planned validation stages defined for Version 1. Together these results establish the first reference implementation of replay-based Executive validation for Internet infrastructure within the GSC-X Platform.

Acknowledgements

The author gratefully acknowledges the maintainers of the Alibaba Cluster Trace Program and the Google Cluster Data project for providing publicly available datasets that enabled independent replay-based validation.

The author also thanks the open-source community whose software, documentation and tools contributed to the development of the GSC-X Platform and the Internet Infrastructure Sandbox.

The author acknowledges the use of OpenAI ChatGPT and Google Gemini as AI-assisted research and writing tools during the preparation of this report. These tools were used to support literature organization, language editing, document structuring, brainstorming and technical review. All research design, software development, validation methodology, experimental implementation, analysis, interpretation of results and final scientific conclusions remain the sole responsibility of the author.

Code and Project Availability

The GSC-X Platform, validation reports and project documentation are publicly available:

- GSC-X GitHub Repository
<https://github.com/sisacek66198068/GSC-X>
- Internet Infrastructure Sandbox
<https://internet.gds-gsc-x.com>

The Internet Infrastructure Sandbox is distributed as part of the broader GSC-X Platform together with validation reports, reproducible replay pipelines and future benchmark updates.

Data Availability

This report is based on publicly available datasets together with reproducible synthetic validation scenarios.

Primary public datasets include:

- Alibaba Cluster Trace Program
- Google Cluster Data

All replay outputs, Executive Benchmark Standard reports and validation artifacts presented in this work were generated within the GSC-X Internet Infrastructure Sandbox.

Version History

Version	Date	Description
V1.0	2026	Initial replay validation framework
V1.0	2026	Alibaba Public Replay
V1.0	2026	Google replay pipeline
V1.0	2026	Public Incident Library
V1.0	2026	Digital Twin prediction
V1.0	2026	What-if analysis
V1.0	2026	Executive Benchmark Standard

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